

# Racial and Ethnic Differences in Breast Cancer Incidence in the United States: A Systematic Review with Quantitative Synthesis

## Abstract

**Background.** In the United States, breast cancer incidence is usually reported for broad racial and ethnic categories. Each category groups several subgroups defined by ancestry, nativity, and region, with receptor-defined subtypes cutting across them, so an aggregate rate may not reflect which subgroups are at highest and lowest risk. Estimates for these subgroups are scattered across separate cancer-registry studies that differ in populations, periods, and reference groups, and are not readily comparable across the published literature.

**Methods.** Following PRISMA 2020 (PROSPERO CRD42023437049), four databases were searched for U.S. studies of female invasive breast cancer incidence by race or ethnicity; those with a recoverable White comparison, usually non-Hispanic White (NHW), were synthesized quantitatively, the rest narratively. Because registries overlap, one representative estimate per group and dimension was selected, not pooled; for American Indian and Alaska Native (AI/AN) populations, an Indian Health Service–linked estimate was preferred to counter registry undercounting. Each was an incidence rate ratio (IRR) versus NHW women, or versus an unstratified White reference in 19 of the 85 cells; ratios not reported directly were recomputed from one master dataset, and risk of bias was assessed with the Joanna Briggs Institute checklist for incidence data.

**Results.** Of 4,793 records screened, 176 studies were included—58 eligible for quantitative synthesis and 118 narrative. Overall incidence was lower than NHW in each aggregate group: Hispanic/Latina 0.72, Asian American, Native Hawaiian, and Pacific Islander (AANHPI) 0.77, AI/AN 0.87, and non-Hispanic Black (NHB) 0.93. Within-group estimates ranged widely—0.16 (Hmong) to 1.27 (Native Hawaiian) across AANHPI subgroups, 0.51 (Mexican) to 0.83 (Puerto Rican) by Hispanic origin, and 0.49 (Navajo area) to 1.33 (Southern Plains) across AI/AN regions. For triple-negative breast cancer, incidence was highest in NHB women (1.95).

**Conclusions.** At the aggregate level, these categories summarize overall differences but not the variation among their subgroups, several of which reach or exceed the NHW rate. This descriptive review compares incidence, not its causes.

## Introduction

Breast cancer is the most commonly diagnosed cancer in U.S. women and a leading cause of cancer death.<sup>1</sup> Its rates are not the same across racial and ethnic groups. In the American Cancer

Society's 2024 report, non-Hispanic Black (NHB) women had breast cancer incidence about 5% lower than non-Hispanic White (NHW) women but 38% higher mortality.<sup>1</sup> NHB women also have the highest rate of triple-negative breast cancer (TNBC),<sup>2</sup> and their tumors, like those in several other minority groups, are diagnosed at younger ages than in NHW women.<sup>3</sup>

Race and ethnicity are recorded in a few broad categories, and rates differ within each. The AANHPI category—Asian American, Native Hawaiian, and Pacific Islander—is a clear example. Treated as one population its incidence sits well below that of NHW women, but that single number covers a wide spread. Using California registry data, Gomez and colleagues<sup>5</sup> reported an incidence of 146 per 100,000 in NHW women and, within Asian American women, a more-than-twofold spread across ethnic groups—from about 103 per 100,000 in Japanese to 46 in Korean women—that widened with US nativity, the rate in US-born Filipina women surpassing that of NHW women. The lowest rates fall among Southeast Asian groups such as the Hmong,<sup>6</sup> and rates differ again across the individual Asian ethnicities followed over time.<sup>7</sup> Hispanic/Latina women are not one group either: incidence differs by country of origin,<sup>8</sup> and among California Hispanic women it varies further with nativity and neighborhood.<sup>9</sup> For American Indian and Alaska Native (AI/AN) women, the aggregate rate also depends on how race is ascertained: registries that assign race by observation tend to misclassify and undercount AI/AN cases, so an unlinked-registry rate can understate their true incidence.<sup>10</sup> Subtype rearranges the order once more: NHW women have the highest rate of HR+/HER2- disease while NHB women have the highest rate of TNBC,<sup>2</sup> so which group ranks highest depends on the tumor being counted.

These estimates are dispersed across separate studies. They are drawn from cancer registries whose populations overlap, and from different periods, age-standardization methods, and reference groups—some studies comparing against White women, others specifically against NHW women. Because many draw on the same or nested registry populations, their estimates are not independent, and combining them directly counts some women more than once. Comparing them therefore calls for gathering the disaggregated estimates across registries, reducing overlapping ones to a single representative, and appraising the quality of what remains.

We reviewed U.S. studies of breast cancer incidence by race and ethnicity and combined them as incidence rate ratios (IRRs) relative to NHW women. The focus was on the disaggregated AANHPI and Hispanic-origin subgroups, on AI/AN women by region, and on the main receptor-defined subtypes. The main analysis retains one representative population-based estimate per analytic cell, and each contributing study was appraised with the Joanna Briggs Institute checklist for studies reporting prevalence and incidence data.

## Methods

### Protocol and reporting

The review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 statement<sup>13</sup> and was registered in PROSPERO (CRD42023437049). The registered protocol pre-specified the research question, the databases and search strategy, the eligibility criteria, the plan to handle overlapping registry data, and the split of eligible studies into a quantitative and a narrative synthesis. Several elements were developed or refined after registration and are reported here as deviations from the protocol: the analytic-cell framework with one representative population-based estimate per cell (in place of statistical pooling of non-independent registry estimates); the provenance tiering and the coverage-based rule for choosing a representative, including the preference for Indian Health Service (IHS)-linked sources for AI/AN populations; the receptor-defined subtype and age-band cells; and the four restriction-based sensitivity analyses.

### Search strategy

Four databases were searched on 7 August 2026: MEDLINE via PubMed, Embase (embase.com Advanced Search), Scopus, and the Web of Science Core Collection. The search combined four concept blocks with AND—breast cancer, race or ethnicity, incidence or age-adjusted rate, and the United States—using database-specific controlled vocabulary together with title/abstract terms, so that race and ethnicity terms were not restricted to the title. Results were limited to 2000–2026, English, and human studies, with document-type exclusions for reviews, letters, editorials, notes, and conference abstracts. The four searches returned 9,099 records. Identification was via these databases only; no registers were searched, and reference lists (citation searching) and grey-literature sources were not used. Full strings, platforms, dates, and per-database counts are given in Supplementary Table 1.

### Eligibility criteria

Studies were eligible for inclusion if they reported invasive breast cancer incidence among women in the United States for at least one racial or ethnic group. Each included study was then assigned to one of two synthesis streams by whether a comparison with a White reference could be recovered. A study entered the **quantitative synthesis** when it expressed incidence relative to a White or NHW comparator—as an IRR or standardized incidence ratio (SIR), or as age-standardized minority and White rates from which a ratio could be computed. A study that met the inclusion criteria but did not yield such a recoverable White comparison—reporting, for example, only age-specific or trend data, rates for a single group, or a reference group that was internal or other than White—was carried in the **narrative synthesis**. Both aggregate groups and

disaggregated subgroups (Asian American, Native Hawaiian and Pacific Islander, Hispanic/Latina by origin, and AI/AN by region) and receptor-defined molecular subtypes were eligible. The reference group was recorded as each study defined it; studies that stratified the reference by Hispanic origin were treated as NHW and those using an unstratified White reference were labelled accordingly, and studies of both kinds were retained because the reference and minority rates came from the same source within each study. The review was limited to U.S.-resident populations. Reviews, editorials, letters, conference abstracts, preprints (not peer-reviewed), non-U.S. studies, reports that did not present invasive female breast cancer incidence by race or ethnicity, and mortality- or survival-only reports were excluded.

A publication that re-reported an estimate already captured from the same registry, diagnosis period, and population was not an independent estimate, so it was never taken as a representative. What then became of it depended on whether another included publication already carried the same content. Where one did, the re-report added nothing and was excluded as a duplicate dataset: 55 reports, of which 51 were earlier editions of the annual cancer-statistics reports or of the Annual Report to the Nation, two were preprint versions of an included analysis, and two were book chapters. Where none did, the report was kept. This applied to the most recent eligible edition of a recurring series, which is the only edition carrying that series' current account; since the registry incidence it reports already came from the dedicated primary study for that registry and period, it contributed to the narrative synthesis alone.

A publication drawing on the same registry family but a different period, region, or subset was a different matter. Such a report gave a distinct estimate and was retained as an overlapping one for the sensitivity analysis rather than excluded; its handling is described under Selection of a representative population-based estimate.

### **Study selection and data extraction**

Records were de-duplicated across databases (4,306 duplicates removed, leaving 4,793 unique records). Titles and abstracts were screened against the eligibility criteria set out above by a large language model (Anthropic's Claude) under the author's direction, working as a single human reviewer with model assistance rather than as two independent reviewers. Records were presented to the model in batches, each showing the title and the abstract truncated to the first 600 characters; the model applied the criteria to each record and proposed an include or exclude decision, and the author reviewed the proposals of each kind. This first pass marked 242 records for full-text retrieval and 4,551 for exclusion under three standardized reasons—not relevant to the research question, not a U.S. population-based or registry study, and editorials, commentaries, letters, or conference abstracts.

Because a single-reviewer screen can miss an eligible study whose abstract omits one of the review's key terms, the excluded records were re-screened with an explicit keyword filter. A record was flagged when its title and abstract together carried at least four of five signal groups: breast cancer (required), an incidence or age-adjusted-rate term (age-standardized rate, IRR, SIR, or per 100,000), a race or ethnicity term, a U.S. context, and a cancer-registry or population-based source. The 183 records carrying all five signals and the 478 missing a single signal were re-read at the full abstract. The re-screen surfaced 17 records for re-adjudication: 14 were reinstated (6 for the quantitative and 8 for the narrative synthesis), 2 were confirmed ineligible on the pre-specified criteria—one on outcome and one on population—and 1 could not be retrieved. Rerouting these 17 records left 4,534 records excluded at screening and 259 reports sought for retrieval.

The author then obtained and read the full texts and made all full-text eligibility and inclusion decisions personally—except for one report available only as a supplementary data table rather than a full article, which was assessed from that table and enters only the narrative synthesis. The model assisted data extraction and cross-checked each extracted value against its source, and every included estimate was verified by the author against the source table, figure, or text. In all, 176 publications were included in the systematic review; of these, 58 were eligible for quantitative synthesis and the remaining 118 informed the narrative synthesis. Most included studies were population-based registry or incidence studies rather than cohort studies (Supplementary Table 2).

These registries form several overlapping systems: the Surveillance, Epidemiology, and End Results (SEER) program collects incidence data from selected states and metropolitan areas; the National Program of Cancer Registries (NPCR) covers registries across the remaining states; United States Cancer Statistics (USCS) pools SEER and NPCR to reach almost the entire U.S. population; the North American Association of Central Cancer Registries (NAACCR) compiles data across member registries; IHS-linked files provide case ascertainment for AI/AN populations; and individual state or regional registries (for example, California and Hawaii) cover single areas.

From each study contributing quantitative data, the cancer registry, geographic coverage, diagnosis period, age range, standard population, racial or ethnic group and comparator, outcome (overall or subtype), the reported estimate, and its confidence interval were extracted. Each estimate was labelled by provenance—directly reported IRR or SIR, or computed from the rates the source reported (with a reported or Poisson-derived variance, or none where neither was available)—and the label was retained for risk-of-bias scoring and for a provenance-restricted

sensitivity analysis. Every estimate was recorded with the minority group as the numerator and the White reference as the denominator. Where a source reported a ratio the other way round, it was inverted to that direction and labelled as inverted rather than recomputed, which happened once: a study that reported its overall Black comparison as NHW over Black while giving its other group comparisons the usual way (Supplementary Note 1).

### **Selection of a representative population-based estimate**

Because SEER and NPCR both feed USCS and many registries report into more than one of these systems, estimates from different publications that draw on the same or nested populations are not statistically independent. Rather than pool such non-independent estimates, the analytic unit was the **analytic cell**—each racial or ethnic group crossed with one analytic dimension (overall incidence, a disaggregated subgroup, a receptor-defined subtype, or an age band)—and one **representative population-based estimate** was selected for each cell. Each estimate was assigned to a registry "family," and the registry family, group, analytic dimension, comparator, diagnosis period, and standard population of each estimate were tabulated so that overlaps within a cell could be seen (Supplementary Table 4). Within a cell, the representative was chosen by applying, in order: the broadest population coverage (USCS > NAACCR > SEER-national > state or regional), then a NHW (rather than an unstratified White) comparator, then the most recent and longest diagnosis period, a clearly documented age-standardization, and a directly reported confidence interval where available. Where a cell had no NHW-comparator source—18 cells, being 16 of the 36 receptor-defined subtype cells and the two Black cells above and below age 40—the estimate using an unstratified White comparator was kept as the representative. One further cell, Alaska Native, also rests on an unstratified White comparator although an NHW-comparator source exists for it, the broader-coverage rule above having selected the registry that reports against an unstratified White reference; 19 of the 85 representatives use such a reference in all. Each estimate's comparator is named in Supplementary Table 4; where such a cell appears in the main text it is marked †.

For AI/AN populations this order was overridden in favour of ascertainment validity: because unlinked registries tend to undercount this population, an estimate with Indian Health Service—based ascertainment—the IHS-PRCDA linkage, or a dedicated registry such as the Alaska Native Tumor Registry—was preferred over an unlinked-registry estimate for the same cell; this ascertainment preference took priority over the coverage and comparator-type criteria above. For the national AI/AN cell this set aside a source covering about 99% of the U.S. population for one confined to Purchased/Referred Care Delivery Area counties; the demoted estimates were carried in the sensitivity analysis. Alaska Native women are an example on the comparator side. The

ascertainment-preferred source—the IHS-eligibility-based Alaska Native Tumor Registry—reports against an unstratified White reference rather than NHW. It was kept as the representative because completeness of ascertainment outweighed comparator type, and the NHW-comparator alternatives were carried in the sensitivity analysis.

Overall, disaggregated-subgroup, subtype, and age-specific results could draw on different studies, but the same registry data were not entered twice for the same question. The stability of the selection was checked in four ways: restricting to studies at low risk of bias, to directly reported (rather than computed) estimates, to estimates with an NHW (rather than unstratified White) comparator, and to estimates standardized to the 2000 U.S. standard population (Supplementary Table 6).

### **Risk of bias**

Risk of bias was assessed with the Joanna Briggs Institute (JBI) critical appraisal checklist for studies reporting prevalence/incidence data,<sup>14</sup> applied to each of the 58 extracted studies by the author (a single assessor) with large-language-model assistance (Supplementary Table 5). Item 9 (survey response rate) was recorded as not applicable, census-like registry ascertainment having no response rate, so each study's rating used the eight applicable items. The checklist records a Yes, No, Unclear or Not applicable for each item and summarises a study as include, exclude or seek further information rather than as a graded rating, so the following rule was set for this review: low risk of bias required at most one "No" with a "Yes" on the two items treated here as key—valid measurement (item 7, including race and ethnicity ascertainment) and appropriate analysis (item 8, age standardization with a variance); three or more "No" ratings gave high risk of bias, and any other combination a moderate rating.

Each study was appraised as the source of the estimates taken from it here rather than as a whole publication, and the items divide accordingly. Items 1 to 7 describe the study and its registry—the sample frame, case ascertainment, case numbers, description of setting and population, coverage, identification of invasive breast cancer, and how race and ethnicity were measured—and were judged at that level, so they do not change with which estimates were extracted. Item 8 depends on the estimates themselves and was judged on those extracted for this review, a variance counting as available when the source printed an interval or when an interval could be recovered from what the source reports.

A study whose extracted estimates carry no interval therefore rates "No" on item 8 whether or not the source printed the ratio itself, which places a point estimate without a variance at moderate rather than low risk of bias. That rating reflects incomplete reporting of uncertainty

rather than evidence that the estimate is biased, and it is the reason several large registry analyses that report ratios without intervals are rated moderate.

Certainty of evidence was not graded, the review being descriptive rather than an estimate of a causal effect.

### **Statistical analysis**

Each estimate was expressed as an IRR relative to NHW women. When a study reported age-standardized rates rather than a ratio, the IRR was computed from the minority and NHW rates of the same study, standard population, and diagnosis period, and its confidence interval was propagated from the reported rate intervals by the delta method when both rates carried an interval, or by scaling the minority-rate interval when the reference rate was a fixed population benchmark without a reported interval. Where a source gave the rates and the underlying case counts but no rate interval, the interval was approximated from the counts on a Poisson assumption ( $SE(\log IRR) = \sqrt{(1/D_{\text{minority}} + 1/D_{\text{NHW}})}$ ,  $D = \text{cases}$ ); this approximation treats the age-standardized rate as though its variance were that of the total case count and so does not fully reflect the age-specific weighting, and it was used only where the source reported neither a rate standard error nor a rate interval (four aggregate estimates and two small single-group rates). Where even the case count was unavailable, the estimate was carried as a point estimate without a confidence interval. Each representative therefore rested on a directly reported ratio or on minority and reference rates from the same source. Two sources reported a standardized incidence ratio (SIR) rather than an IRR—South Asian women and Alaska Native women, each against the US White population; because an SIR indirectly standardizes to the reference population's age structure rather than expressing two directly standardized rates on a common standard, both were read on the same relative-to-White scale but labelled as SIRs (Supplementary Table 4) and excluded from the directly-reported-IRR sensitivity analysis. Neither is the representative for its cell, so no estimate reported in the main text is an SIR. The standard population was recorded as each source stated it—2000 US for most, with a 1970 world or 1970 US standard in a few older studies and the 1960 Segi world standard in one; SEER\*Stat and USCS age-adjusted rates use the 2000 US standard by default, and that default was recorded where a study reported rates from those programs without naming a standard. Where a source neither stated a standard population nor produced its rates through such a program (one small-area Bayesian analysis), the standard was recorded as not stated rather than inferred from the size of the rates. One estimate paired an Alaska Native rate with an external SEER-Explorer NHW reference<sup>64</sup>, its source having reported no in-paper White comparator; it was retained only as an overlap record, not as a representative. For the non-independence reason given above, these

estimates were not combined into a pooled random-effects estimate; the racial and ethnic differences and the subgroup heterogeneity are presented group by group rather than as a single pooled value. Studies that met the inclusion criteria but did not provide a recoverable NHW comparison were not placed on the IRR scale; these were summarized narratively by racial or ethnic group and outcome. Analyses were carried out in Python 3 using SciPy.

All quantitative results were generated from a single master extraction dataset, from which the main-text table, the forest figures, and the sensitivity tables were regenerated.

## **Results**

### **Selection and characteristics of included studies**

The four databases returned 9,099 records; 4,306 duplicates were removed, leaving 4,793 unique records for title and abstract screening. Full text was sought for 259 reports; 10 could not be retrieved, and of the 249 assessed, 73 were excluded (reasons in Supplementary Table 3), giving 176 included studies (Figure 1). Of these, 58 were eligible for quantitative synthesis and entered the analysis, contributing 218 individual estimates across racial/ethnic groups and analytic dimensions<sup>2–9,11–12,15–63</sup>—22 supplied a cell representative, and the remainder contributed overlapping estimates carried only in the sensitivity analysis; the remaining 118 informed the narrative synthesis ( $176 = 58 + 118$ ). The extracted studies were published between 2002 and 2026 and drew on SEER, NPCR and USCS, NAACCR, IHS-linked and tribal registries, and individual state or regional registries (Supplementary Table 2). After collapsing overlapping registry-family estimates, 85 representative estimates remained, one per analytic cell; all are listed in Supplementary Table 4, and the 38 covering the principal comparisons—overall incidence, the disaggregated AANHPI and Hispanic-origin subgroups, AI/AN by region, Middle Eastern populations, and TNBC—are shown in Table 1; the remaining cells, chiefly the other receptor-defined subtypes and the age-specific comparisons, are in the Supplementary Materials. The reference group was NHW for 66 of the 85 representatives and an unstratified White group for the other 19; five of those are among the cells shown in Table 1, where they carry a †, and Supplementary Table 4 names the comparator of every extracted estimate in a column rather than by mark.

### **Overall invasive breast cancer incidence**

Relative to NHW women, overall invasive breast cancer incidence was lower in each aggregate minority group: Hispanic/Latina women IRR 0.72 (95% CI 0.71–0.73), the AANHPI aggregate 0.77 (0.75–0.79), AI/AN women 0.87 (a point estimate; the Indian Health Service–linked

national source reported no confidence interval), and NHB women 0.93 (0.92–0.95) (Table 1; Figure 2).

### **Disaggregated AANHPI subgroups**

Within the AANHPI aggregate, incidence varied across Asian American subgroups from 0.16 (0.12–0.22) in Hmong women to 1.05 (1.01–1.08) in Japanese women, with Cambodian 0.26 (0.20–0.34), Laotian 0.39 (0.35–0.43), Vietnamese 0.59 (0.57–0.61), Korean 0.68 (0.65–0.70), Chinese 0.76 (0.75–0.78), Filipina 0.86 (0.84–0.87), and Asian Indian/Pakistani 0.96 (0.94–0.98) in between; the Asian American aggregate was 0.79 (0.78–0.80). Among Native Hawaiian and Pacific Islander (NHPI) subgroups, Guamanian/Chamorro/Samoan women had an IRR of 0.72 (0.64–0.82) and Native Hawaiian women 1.27 (1.21–1.33), with an NHPI aggregate of 1.21 (1.17–1.26) (Table 1; Figure 2).

### **Hispanic/Latina by country of origin**

Among Hispanic/Latina women, incidence in Florida differed by country of origin: Mexican 0.51 (0.38–0.69), New Latino 0.70 (0.63–0.77), Cuban 0.77 (0.69–0.86), and Puerto Rican 0.83 (0.74–0.94), against an aggregate Hispanic/Latina IRR of 0.72 (Table 1; Figure 2).

### **AI/AN by region**

AI/AN incidence relative to NHW women varied widely by region. Across the six Indian Health Service regions the IRR ranged from 0.57 in the Southwest and 0.61 in the East, through 0.93 in the Pacific Coast and 1.05 in the Northern Plains, to 1.33 (1.26–1.41) in the Southern Plains; the tribal-specific Navajo area was lowest at 0.49 (0.44–0.55). These regional figures and the national AI/AN aggregate (0.87) come from Indian Health Service–linked sources, which correct the undercounting of AI/AN cases in unlinked registries; they are point estimates without a reported confidence interval, except the Navajo area, Alaska Native, and Southern Plains. Alaska Native women, ascertained by the IHS–eligibility–based Alaska Native Tumor Registry, had an IRR of 1.09 (0.99–1.21) against an unstratified White reference (Table 1; Figure 2).

### **Middle Eastern populations**

Women identified as Middle Eastern had an IRR of 0.86 (0.84–0.88) relative to NHW women (Table 1).

### **Molecular subtypes**

For TNBC, incidence relative to NHW women was highest in NHB women (1.95, 1.93–1.98). The AANHPI aggregate was 0.70 (0.68–0.72), Hispanic 0.86 (0.84–0.88), and AI/AN 0.86 (0.80–0.93); within AANHPI, TNBC IRRs were lower in Chinese (0.53 †), Filipina (0.84 †), and

Native Hawaiian (0.86 †) women (Table 1). Across groups and analytic dimensions the relative ordering shifts with the outcome examined (Figure 3): NHB women, close to the NHW rate for overall incidence (0.93), have the highest triple-negative rate (1.95), while several Asian American subgroups that are below the NHW rate overall remain so across subtypes. Estimates for the other receptor-defined subtypes (HR+/HER2–, HR+/HER2+, and HR–/HER2+) rest mostly on single unstratified-White sources and are listed in Supplementary Table 4; they are the receptor-subtype cells that the sensitivity analyses show to be least stable.

### **Narrative synthesis**

The 118 narrative-synthesis studies met the inclusion criteria but contributed no extractable quantitative estimate—most because no IRR or NHW-referenced age-standardized rate could be recovered, and a few multi-registry summary reports (for example, the annual reports to the nation and the annual cancer-statistics summaries) because they re-reported registry incidence already quantified from the dedicated primary study for the same registry and period. Grouped by theme in Supplementary Table 2, they fall into six overlapping topics, and where they reported a racial or ethnic comparison their findings were broadly consistent with the quantitative results.

Molecular subtype and histology (28 studies) described the higher triple-negative and hormone-receptor-negative burden in NHB women and the predominance of HR+/HER2– disease in NHW women, and extended the picture to inflammatory, lobular, and ductal histologies that the receptor-defined cells do not capture. Geography and region (24 studies) described variation in incidence across states, counties, and rural–urban gradients and within particular registries, including Delta-region, Appalachian, and single-state analyses. Age and early-onset (20 studies) reported racial differences concentrated in younger women, among them the Black–White age crossover and a higher early-onset burden in NHB women. Socioeconomic status and screening (9 studies) tied incidence differences to poverty, neighborhood deprivation, insurance, and mammography use rather than to race alone. Nativity and immigrant generation (8 studies) reported incidence rising with longer US residence and across immigrant generations in Asian, Hispanic, and Middle Eastern populations, echoing the nativity and enclave gradients seen in the quantitative subgroups. Time trends (8 studies) tracked rising incidence among Asian American women and falling estrogen-receptor-negative rates over recent decades. The remaining 21 were subgroup-specific descriptive reports—single Asian ethnicities or AI/AN young adults, for example—without a recoverable NHW comparison.

Taken together these narrative findings reproduce the main quantitative patterns—lower overall incidence in most minority groups than in NHW women, wide variation among Asian American

and Hispanic/Latina subgroups, and a higher triple-negative burden in NHB women. Because the studies used varied reference groups, standard populations, and reporting formats, their results were summarized qualitatively rather than placed on the common IRR scale.

### **Risk of bias and sensitivity analyses**

On the JBI checklist, 41 of the 58 studies were at low risk of bias and 17 at moderate risk, with none at high risk (Supplementary Table 5); the moderate ratings arose mainly where an estimate was reported without a variance, the standard population was not clearly specified, or race and ethnicity ascertainment was limited. The representative selection was examined in four ways (Supplementary Table 6), and the pattern held throughout. Restricting to low-risk-of-bias studies left 54 of 85 cell representatives unchanged, with the 19 changed and 12 dropped cells falling mainly among the disaggregated AANHPI subgroups and the AI/AN cells, and in smaller numbers among the receptor-subtype, age-specific, and Middle Eastern cells; the largest change was the AI/AN national aggregate, where the restriction replaced the 2010-2015 IHS-linked estimate (0.87) with the 1999-2004 IHS-linked estimate (0.63) (Supplementary Table 6a); restricting to directly reported IRRs—which also excludes the two indirectly standardized SIRs—left 35 unchanged (4 changed, 46 dropped; Supplementary Table 6b); restricting to NHW-comparator estimates left 66 unchanged (1 changed, 18 dropped; Supplementary Table 6c), the 18 dropped cells being those whose only representative used an unstratified White comparator—16 receptor-defined subtype cells and the two Black cells above and below age 40—and the one changed cell being Alaska Native, whose unstratified-White representative (1.09) gave way to an Indian Health Service-linked estimate reported against an NHW reference (1.25); and restricting to the 2000 U.S. standard population changed little, since 83 of the 85 representatives were already standardized to it, so the restriction dropped no cell and changed only two Black receptor-subtype cells, each from a 1960 Segi-world-standardized representative to a 2000 U.S.-standardized estimate in the same direction (hormone-receptor-negative 1.80 to 1.60, hormone-receptor-positive 0.82 to 0.70; Supplementary Table 6d).

### **Discussion**

This review assembled disaggregated, population-based estimates of invasive breast cancer incidence for U.S. racial and ethnic groups against a common NHW comparator (an unstratified White reference for 19 of the 85 cells, most of them receptor-defined subtypes). At the aggregate level, each minority group had lower overall incidence than NHW women—Hispanic/Latina women had the lowest IRR (0.72), followed by AANHPI (0.77) and AI/AN (0.87) women, with NHB women closest to NHW (0.93). For TNBC, however, NHB women had the highest incidence (1.95), while most other groups were at or below the NHW rate.

The central observation is that these aggregate figures average over the wide variation among the subgroups they contain. Among Asian American subgroups the IRR ranged from 0.16 in Hmong women to 1.05 in Japanese women; Native Hawaiian women (1.27) and the NHPI aggregate (1.21) exceeded the NHW rate; Hispanic incidence rose from 0.51 in Mexican women to 0.83 in Puerto Rican women; and AI/AN incidence ranged from 0.49 in the Navajo area to 1.33 in the Southern Plains, against a national AI/AN aggregate of 0.87 (Figure 2; each aggregate and subgroup is the representative estimate from a separate study placed on the common NHW scale). This pattern is consistent with earlier disaggregated work and extends it: Gomez and colleagues first documented the variation among Asian American subgroups,<sup>5</sup> which the present estimates span over the most recent SEER period;<sup>31</sup> the high burden in Native Hawaiian women echoes reports from Hawaii;<sup>24</sup> the Hispanic gradient reproduces that described among first-generation U.S. Hispanics;<sup>8</sup> the AI/AN regional range follows the pattern in earlier Indian Health Service–linked regional reports,<sup>10,53</sup> and the low Navajo-area rate that in the Navajo Nation registry analysis;<sup>48</sup> and the higher TNBC incidence in NHB women is concordant with prior subtype-specific work.<sup>2,23</sup> The relative ordering of groups also shifts with the outcome examined (Figure 3).

For the subgroups whose incidence sits close to the NHW rate, which study represents the cell can change the direction of the small difference. Japanese women are the clearest example: the representative estimate, from a recent SEER source,<sup>31</sup> places their overall incidence just above the NHW rate (1.05), but that study is at moderate risk of bias, and when the representative selection is restricted to low-risk-of-bias studies the cell is instead filled by an earlier eight-state SEER+NPCR estimate<sup>60</sup> that places it just below (0.95). The flip does not overturn the finding that Japanese incidence is near the NHW level, but it shows that the sign of a near-null difference is not robust for this subgroup. Such a change on restriction reflects more than risk of bias: the substituted study also differs in diagnosis period (2009–2011 versus 2018–2022) and in the registries it draws on, either of which can move a borderline estimate across 1.0.

Two features of the underlying data shaped these estimates: how AI/AN cases were ascertained, and which White group each source compared against. Unlinked registries tend to undercount AI/AN cases, so the AI/AN estimates here draw on Indian Health Service–linked sources, which capture these cases more completely.<sup>10,53</sup> The two national figures differ substantially—0.56 from an unlinked source and 0.87 from an IHS-linked one—but they come from separate studies, so the gap cannot be read as the effect of linkage alone. Three differences run together. The unlinked estimate covers 1999–2015 and the IHS-linked one 2010–2015, and the AI/AN ratio rose over that interval. The IHS-linked figure is confined to Purchased/Referred Care Delivery Area counties, and that restriction applies to the NHW comparison group as well, so its

numerator and denominator both describe a different population from the national one. Those counties contain or adjoin federally recognized tribal lands and hold about 53% of the U.S. AI/AN population, with regional coverage from 16% in the East to all of Alaska,<sup>53</sup> so the IHS-linked figures describe about half of this population rather than the whole of it, and the better-covered regions are those where AI/AN residents make up a larger share of the county population. Only the third difference—how AI/AN race was ascertained—is the linkage itself.

Among IHS-linked national estimates the ratio also rose over time, from 0.63 in 1999–2004 to 0.87 in 2010–2015, so the most recent IHS-linked estimate was taken as the representative and the earlier one retained for sensitivity. The IHS-linked regional estimates vary widely and reach or exceed the NHW rate in the Northern and Southern Plains—and, against an unstratified White reference, among Alaska Native women (regional values in Table 1 and Figure 2). The lower unlinked values therefore likely reflect incomplete case ascertainment, together with the wider diagnosis periods and the national rather than PRCDA coverage of those sources, rather than only lower underlying incidence; the AI/AN–NHW comparison therefore depends on the source's ascertainment as well as its diagnosis period and geographic scope.<sup>41,58</sup> The IHS-linked correction, however, was available only for the aggregate and regional estimates: no Indian Health Service–linked source reported AI/AN incidence by molecular subtype, so the AI/AN subtype figures (for example, the HR-positive/HER2-negative and triple-negative cells) rest on unlinked registries and may understate the true rates.

Reference groups also differed across sources—66 of the 85 representatives used an NHW comparator and 19 an unstratified White reference—so each comparator was recorded as its source defined it and a sensitivity analysis was restricted to NHW-comparator estimates; the unstratified-White cells, 16 of them receptor-defined subtypes, can shift a ratio slightly.

These cell estimates are not fully commensurable, for the reasons set out under the limitations below, so their arrangement on the common NHW scale shows the span of published figures rather than a set of mutually calibrated rates. The main pattern held for most of the cells each restriction could re-examine, but not uniformly. Cells that no estimate could fill under a restriction were dropped, and a dropped cell is untested under that restriction rather than confirmed by it—the directly-reported restriction alone leaves 46 of the 85 cells unexamined, because registries publish rates rather than ratios. Standard-population differences changed little. The low-risk-of-bias restriction moved the most cells (19 of the 73 it could re-examine), concentrated in the disaggregated AANHPI subgroups, the age-specific cells whose estimates sit near the NHW rate, as with the Japanese cell above, and the AI/AN cells. The AI/AN change is the largest, and it turns on how uncertainty was reported rather than on a difference in bias. The

national aggregate's representative—the 2010–2015 IHS-linked estimate of 0.87—is rated moderate only because it prints no confidence interval, so the restriction replaces it with the 1999–2004 IHS-linked estimate of 0.63, which does print intervals. The gap between the two is the rise over time described above, not two sources disagreeing about one period. That substitution nonetheless moves AI/AN below Hispanic/Latina women, so the ordering of the aggregates is not preserved under that restriction, and neither the ordering nor a few near-null subgroup signs should be read as established.

The review is descriptive and does not identify causes. Several explanations proposed in the included studies are best treated as possible contributors rather than established causes: nativity and generational status are the most consistently reported, with lower incidence in some Asian and Hispanic subgroups linked to a larger foreign-born proportion and residence in ethnic enclaves and to rates that rise with longer U.S. residence,<sup>5,9</sup> and differences in molecular-subtype distribution and in screening and access to care have also been described.<sup>2,34</sup> Genetic or biological mechanisms were not examined.

The review has limitations. About three fifths of the representative rate ratios (50 of 85) were computed from published age-standardized rates rather than reported directly, which can move a ratio slightly. Expressing each group's incidence as a ratio to the NHW rate does not by itself make the estimates commensurable. Within a cell the numerator and denominator share one standard population, period, and region, so the ratio is internally consistent; the ratio does not, however, cancel the standard population, because the age-specific rates of two groups differ in shape and another set of standard weights would move their standardized rates—and so their ratio—by different amounts. Across cells, differences in standard population, underlying age structure, registry region, and diagnosis period therefore remain embedded. Diagnosis period and registry region are the main limits on cross-cell comparability; standard population differed less—most estimates used the 2000 US standard, with a 1970 (world or US) or the 1960 Segi world standard in a few older studies, and restricting to the 2000 US standard left the pattern essentially unchanged. The review was restricted to U.S.-resident populations, so the estimates are not generalizable to countries that use a different White reference, different racial and ethnic categories, and different standard populations. Finally, estimates for the smallest subgroups rested on limited numbers, studies contributing only to the narrative synthesis were not quantified, and grey literature was not searched. Screening, full-text selection, and extraction were carried out by a single reviewer with large-language-model assistance rather than by two independent reviewers. To limit studies missed by this single-reviewer screening, the excluded records were re-screened at the full abstract with an explicit keyword filter, leaving the residual

risk of a missed study confined to reports whose abstracts omit two or more of the screening signals.

What this review adds beyond the individual studies is threefold. First, it places estimates that were scattered across separate registry reports—each with its own reference group, standard population, and diagnosis period—onto a single NHW-referenced scale, so the disaggregated subgroups and their aggregates can be seen together in one place. Second, by collapsing overlapping registry-family estimates while retaining the overlaps for the sensitivity analysis, it separates the cells that more than one study could fill (38 of the 85 analytic cells, and in 27 of those the studies come from at least two different registry families) from those resting on a single study (47). These sources are not fully independent—US registries are nested, so publications from different families can still draw on overlapping cases—but where more than one is available they agree in most cells, diverging chiefly in the near-null AANHPI subgroups and the ascertainment-dependent AI/AN comparison. Third, it makes the evidence gaps explicit: Middle Eastern women, several Native Hawaiian and Pacific Islander and Hispanic-origin subgroups, and AI/AN incidence by molecular subtype each rest on a single regional or unlinked source, and the AI/AN regional and national aggregate figures are point estimates without a confidence interval—groups for which a dedicated, and for AI/AN an IHS-linked, primary study is still needed. Because studies differed in diagnosis period and region, these estimates are best read as a comparable-scale summary of the available evidence, with the provenance and comparator of each recorded—marked in the main-text table and figures (computed CI ‡; unstratified White †) and named in full in Supplementary Table 4—so readers can weigh them.

In summary, aggregate racial and ethnic categories describe broad differences in U.S. breast cancer incidence but do not capture the range within them, and the subtype pattern for NHB women departs from the overall pattern. Each comparison rests on a single appraised representative estimate per analytic cell rather than a pooled figure.

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## Tables

**Table 1. Incidence rate ratios of invasive breast cancer among U.S. racial and ethnic groups relative to non-Hispanic White women, by analytic dimension**

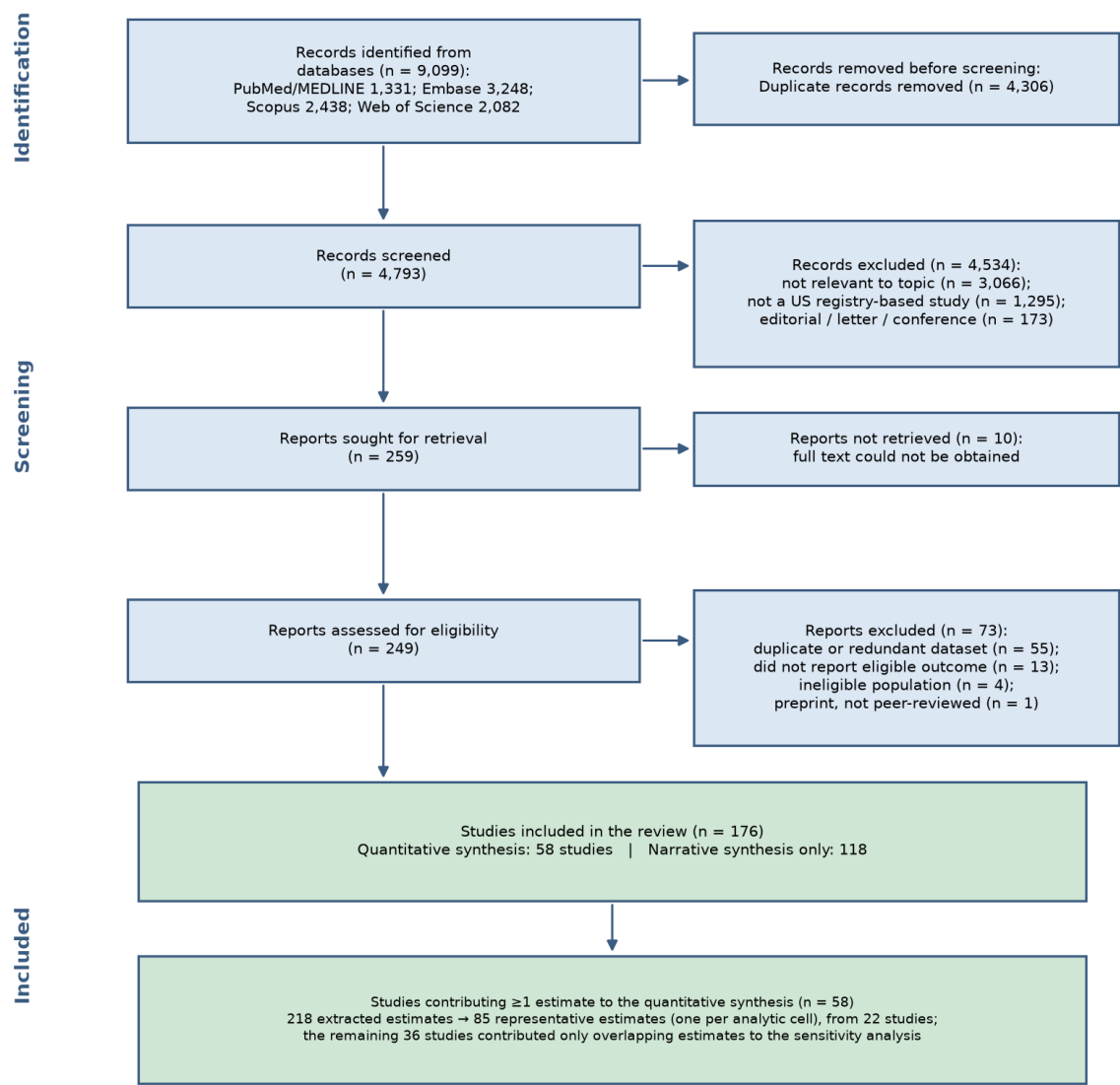
| Group                                                          | Estimate [95% CI]      | Representative study           | Registry family                           | RoB      |
|----------------------------------------------------------------|------------------------|--------------------------------|-------------------------------------------|----------|
| <b>Overall invasive breast cancer</b>                          |                        |                                |                                           |          |
| Hispanic                                                       | 0.718 [0.707, 0.731] ‡ | Ellington 2022[25] (2018)      | USCS(NPCR+SEER ~99%)                      | Low      |
| Asian American, Native Hawaiian, and Pacific Islander (AANHPI) | 0.769 [0.752, 0.788] ‡ | Ellington 2022[25] (2018)      | USCS(NPCR+SEER ~99%)                      | Low      |
| American Indian and Alaska Native (AI/AN)                      | 0.87 (point est.)      | Melkonian 2019[53] (2010-2015) | IHS-PRCDA                                 | Moderate |
| non-Hispanic Black (NHB)                                       | 0.933 [0.920, 0.946] ‡ | Ellington 2022[25] (2018)      | USCS(NPCR+SEER ~99%)                      | Low      |
| <b>Asian American subgroups</b>                                |                        |                                |                                           |          |
| Hmong                                                          | 0.164 [0.120, 0.224] ‡ | Mills 2005[6] (1988-2000)      | California-CCR                            | Low      |
| Cambodian                                                      | 0.264 [0.204, 0.341] ‡ | Kem 2007[55] (1998-2002)       | California-CCR                            | Moderate |
| Laotian/Kampuchean                                             | 0.388 [0.350, 0.430] ‡ | Gomez 2026[31] (2018-2022)     | SEER-national                             | Moderate |
| Vietnamese                                                     | 0.586 [0.566, 0.608] ‡ | Gomez 2026[31] (2018-2022)     | SEER-national                             | Moderate |
| Korean                                                         | 0.677 [0.654, 0.701] ‡ | Gomez 2026[31] (2018-2022)     | SEER-national                             | Moderate |
| Chinese                                                        | 0.760 [0.745, 0.775] ‡ | Gomez 2026[31] (2018-2022)     | SEER-national                             | Moderate |
| Filipina                                                       | 0.857 [0.839, 0.874] ‡ | Gomez 2026[31] (2018-2022)     | SEER-national                             | Moderate |
| Asian Indian/Pakistani                                         | 0.958 [0.937, 0.979] ‡ | Gomez 2026[31] (2018-2022)     | SEER-national                             | Moderate |
| Japanese                                                       | 1.045 [1.009, 1.083] ‡ | Gomez 2026[31] (2018-2022)     | SEER-national                             | Moderate |
| Asian American (aggregate)                                     | 0.792 [0.784, 0.800] ‡ | Gomez 2026[31] (2018-2022)     | SEER-national                             | Moderate |
| <b>Native Hawaiian and Pacific Islander (NHPI) subgroups</b>   |                        |                                |                                           |          |
| Guamanian/Chamorro/Samoan                                      | 0.723 [0.641, 0.817] ‡ | Gomez 2026[31] (2018-2022)     | SEER-national                             | Moderate |
| Native Hawaiian                                                | 1.270 [1.213, 1.330] ‡ | Gomez 2026[31] (2018-2022)     | SEER-national                             | Moderate |
| Native Hawaiian and Pacific Islander (NHPI)                    | 1.212 [1.168, 1.258] ‡ | Gomez 2026[31] (2018-2022)     | SEER-national                             | Moderate |
| <b>Hispanic/Latina by country of origin</b>                    |                        |                                |                                           |          |
| Mexican                                                        | 0.512 [0.382, 0.686] ‡ | Pinheiro 2009[8] (1999-2001)   | State: Florida                            | Low      |
| New Latino                                                     | 0.697 [0.629, 0.771] ‡ | Pinheiro 2009[8] (1999-2001)   | State: Florida                            | Low      |
| Cuban                                                          | 0.769 [0.688, 0.860] ‡ | Pinheiro 2009[8] (1999-2001)   | State: Florida                            | Low      |
| Puerto Rican                                                   | 0.833 [0.738, 0.939] ‡ | Pinheiro 2009[8] (1999-2001)   | State: Florida                            | Low      |
| <b>American Indian and Alaska Native (AI/AN) by region</b>     |                        |                                |                                           |          |
| AI/AN (Navajo)                                                 | 0.49 [0.44, 0.55]      | Yazzie 2025[48] (2014-2018)    | IHS-PRCDA                                 | Low      |
| AI/AN (Southwest)                                              | 0.57 (point est.)      | Melkonian 2019[53] (2010-2015) | IHS-PRCDA                                 | Moderate |
| AI/AN (East)                                                   | 0.61 (point est.)      | Melkonian 2019[53] (2010-2015) | IHS-PRCDA                                 | Moderate |
| AI/AN (Pacific Coast)                                          | 0.93 (point est.)      | Melkonian 2019[53] (2010-2015) | IHS-PRCDA                                 | Moderate |
| AI/AN (Northern Plains)                                        | 1.05 (point est.)      | Melkonian 2019[53] (2010-2015) | IHS-PRCDA                                 | Moderate |
| Alaska Native                                                  | 1.09 [0.99, 1.21] †    | Nash 2019[11] (2009-2014)      | Alaska Native Tumor Registry (IHS-linked) | Low      |
| AI/AN (Southern Plains)                                        | 1.33 [1.26, 1.41]      | Melkonian 2021[58] (2012-2016) | IHS-PRCDA                                 | Low      |
| <b>Middle Eastern</b>                                          |                        |                                |                                           |          |
| Middle Eastern                                                 | 0.86 [0.84, 0.88]      | Nasseri 2009[46] (1988-2004)   | California-CCR                            | Moderate |

| Group                                                          | Estimate [95% CI]     | Representative study       | Registry family              | RoB |
|----------------------------------------------------------------|-----------------------|----------------------------|------------------------------|-----|
| <b>Triple-negative breast cancer</b>                           |                       |                            |                              |     |
| Chinese                                                        | 0.53 [0.31, 0.93] † ‡ | Loo 2019[24] (2010-2013)   | Hawaii Tumor Registry (SEER) | Low |
| Asian American (aggregate)                                     | 0.61 [0.58, 0.65]     | Gomez 2017[35] (2009-2013) | California-CCR               | Low |
| Asian American, Native Hawaiian, and Pacific Islander (AANHPI) | 0.700 [0.680, 0.720]  | Sung 2023[23] (2015-2019)  | USCS(NPCR+SEER ~99%)         | Low |
| Filipina                                                       | 0.84 [0.60, 1.18] † ‡ | Loo 2019[24] (2010-2013)   | Hawaii Tumor Registry (SEER) | Low |
| American Indian and Alaska Native (AI/AN)                      | 0.860 [0.800, 0.930]  | Sung 2023[23] (2015-2019)  | USCS(NPCR+SEER ~99%)         | Low |
| Hispanic                                                       | 0.860 [0.840, 0.880]  | Sung 2023[23] (2015-2019)  | USCS(NPCR+SEER ~99%)         | Low |
| Native Hawaiian                                                | 0.86 [0.60, 1.22] † ‡ | Loo 2019[24] (2010-2013)   | Hawaii Tumor Registry (SEER) | Low |
| Japanese                                                       | 1.08 [0.79, 1.46] † ‡ | Loo 2019[24] (2010-2013)   | Hawaii Tumor Registry (SEER) | Low |
| non-Hispanic Black (NHB)                                       | 1.950 [1.930, 1.980]  | Sung 2023[23] (2015-2019)  | USCS(NPCR+SEER ~99%)         | Low |

*Note. This table shows 38 of the 85 analytic cells, those covering the principal comparisons; the rest, chiefly the other receptor-defined subtypes and the age-specific comparisons, are in Supplementary Table 4. Each cell shows one representative population-based estimate — the most recent, broadest-coverage registry estimate per analytic cell — not a pooled estimate; the selection rule and its robustness are given in the Methods and Supplementary Table 6. Each representative study is followed by its number in the reference list. The effect measure is the incidence rate ratio (IRR) throughout; comparisons are versus non-Hispanic White (NHW) women. † the study's reference was an unstratified White group (not stratified by Hispanic origin), examined in the NHW-comparator sensitivity analysis (Supplementary Table 6c); 5 of the 19 cells so marked appear in this table. ‡ the 95% confidence interval was computed by the reviewers from published rates rather than reported in the source. NHB, non-Hispanic Black; AANHPI, Asian American, Native Hawaiian, and Pacific Islander (NHPI, the Pacific Islander subset, shown separately); RoB, risk-of-bias rating of the representative study. Full per-estimate detail is in the Supplementary Materials.*

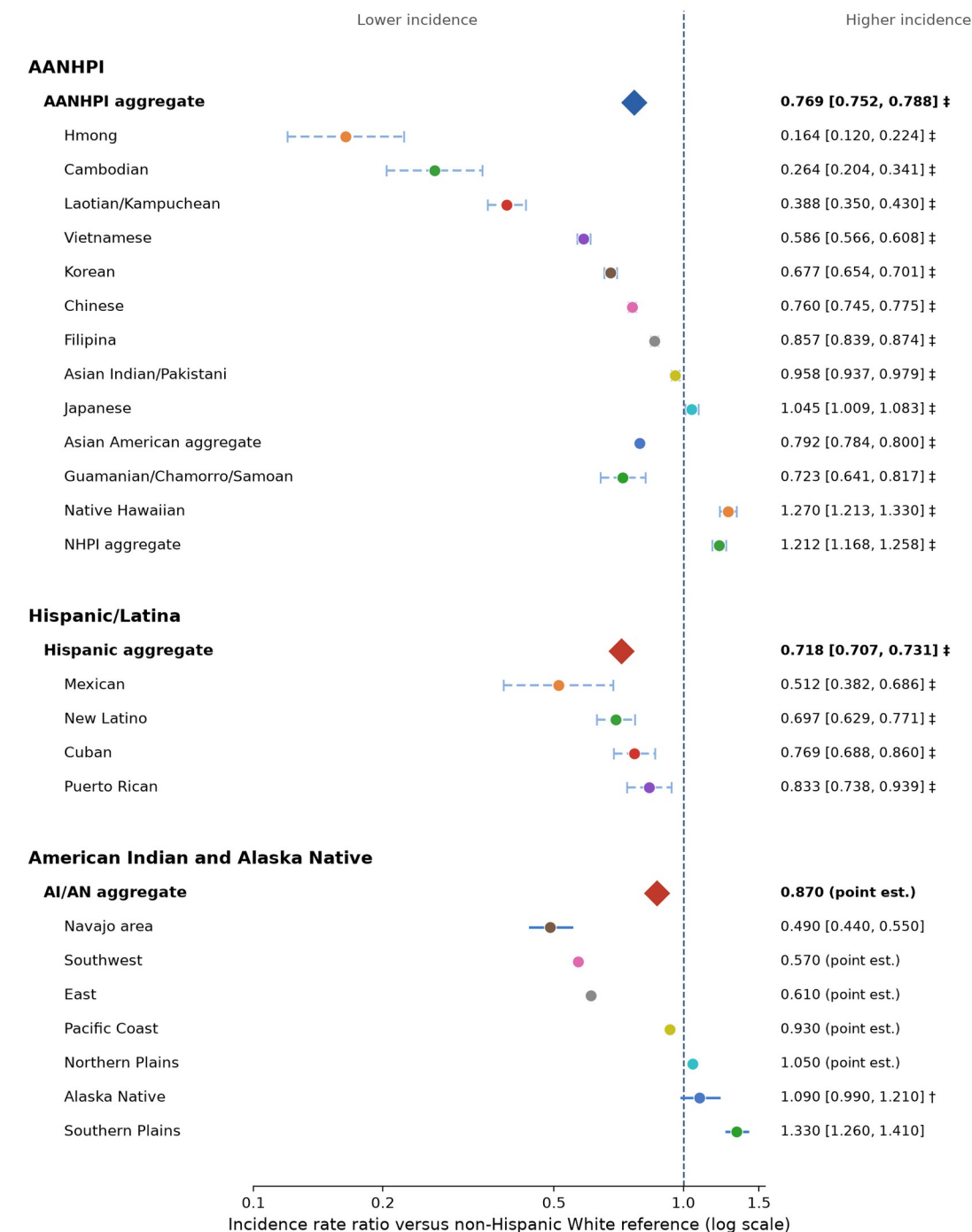
Figures

Figure 1. PRISMA 2020 flow diagram



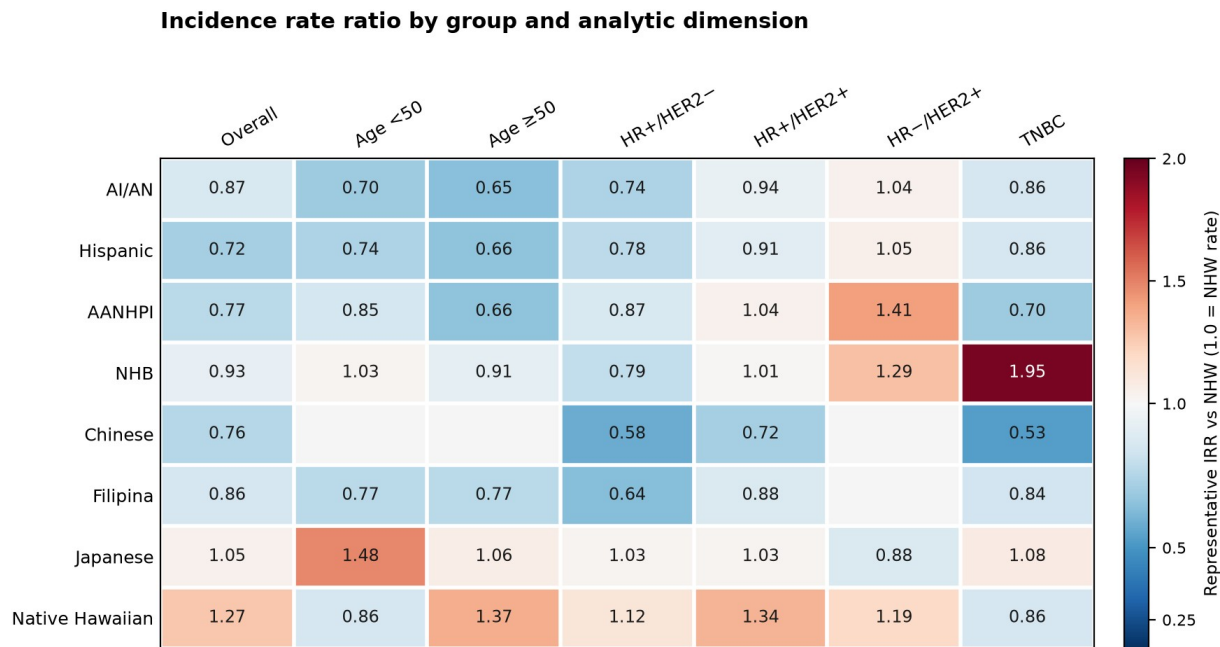
PRISMA 2020 flow diagram. Identification via databases only; no registers, citation searching, or grey-literature sources were used. Single-reviewer screening with AI assistance. Search date 7 August 2026.

**Figure 2. Aggregate-to-disaggregated heterogeneity in breast cancer incidence (representative IRR vs non-Hispanic White, 95% CI): the Asian American, Native Hawaiian and Pacific Islander (AANHPI), Hispanic/Latina, and American Indian and Alaska Native (AI/AN) aggregates, each shown with its subgroups**



*Note. Each point is the representative estimate for one analytic cell, drawn from a separate study; an aggregate and its subgroups differ in registry, region, diagnosis period, and standard population. Diamonds mark aggregate groups, circles subgroups. A solid bar is a 95% CI reported in the source; a dashed bar marked ‡ is one computed for this review; a point without a bar had no interval in its source. † reference was an unstratified White group, not NHW. Values match Table 1.*

**Figure 3. Incidence rate ratio by racial or ethnic group and analytic dimension (representative estimate vs non-Hispanic White; 1.0 = the NHW rate)**



Blank (grey) cells: no representative estimate for that group × dimension. Subgroup rows show Asian American groups within the AANHPI aggregate.